Lecture-5

Congruences.

Def^n - Let a, b, n be integers with $n \neq 0$.

$$a \equiv b \pmod{n}$$

(a is congruent to b mod n)
if $a-b$ is a multiple (Positive or Negative) of n.

$\Rightarrow a \equiv b \pmod{n}$ if a and b differ by a multiple of n. So $a - b = nk$ or $a = b + nk$

Ex. $32 \equiv 2 \pmod{5}$, $-12 \equiv 37 \pmod{7}$

$17 \equiv 17 \pmod{13}$

Proposition - Let a, b, c, n be integers with $n \neq 0$.

(i) $a \equiv 0 \pmod{n}$ if and only if $n \mid a$.

(ii) $a \equiv a \pmod{n}$.

(iii) $a \equiv b \pmod{n}$ if and only if $b \equiv a \pmod{n}$

(iv) If $a \equiv b$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$

Sol: (i) $a \equiv 0 \pmod{n}$ means $a = a - 0$ is a multiple of n, which is $n \mid a$.

(ii) $a - a = 0 \cdot n$, $k = 0$

(iii) $b \equiv a \mod n \Rightarrow b - a = nk \Rightarrow a - b = n(-k)$

So $a \equiv b \pmod{n}$

(iv) $a = b + nk$ and $b = c + l$ So

$a = c + l + nk \Rightarrow a = c + (l+k)n \Rightarrow a \equiv c \pmod{n}$

mod operation on integers

We will work with integers mod n, denoted by Z_n.

These may be regarded as the set $\{0, 1, 2, \ldots n-1\}$, with addition, subtraction and multiplication mod n.

If integer a is divided by n with remainder r then

$$a = nq + r \quad \text{with} \quad 0 \le r < n.$$

So $a \equiv r \pmod n$ So every a is congruent to integer r mod n.

Proposition. Let a, b, c, d, n be integers with $n \neq 0$, and $a \equiv b \pmod n$ and $c \equiv d \pmod n$ then

$a+c \equiv b+d$, $a-c \equiv b-d$, $ac \equiv bd \pmod n$

Sol. $a = b + nk$, $c = d + nl$ then $a+c = b+d + n(k+l)$ etc.

Addition over modulo n -

Let $n = 6$ the addition modulo 6 table is -

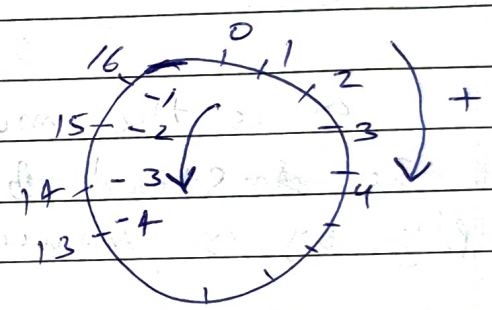
+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Table for Multiplication mod 6 -

X	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Ex. Solve $x + 7 \equiv 3 \pmod{17}$

Sol. $x \equiv (3-7) \bmod n \equiv -4 \bmod 17 \equiv 13 \bmod 17$



Proposition- Let a, b, c, n be integers with $n \neq 0$ and with $\gcd(a, n) = 1$.
If $ab \equiv ac \pmod{n}$, then $b \equiv c \pmod{n}$.
In other words, if a and n are relatively prime, we can divide both sides of the congruence by a.

Ex. Solve $2x + 7 \equiv 3 \pmod{17}$

$2x \equiv 3 - 7 \pmod{17} \Rightarrow 2x \equiv -4 \pmod{17}$

$x \equiv -2 \pmod{17} \Rightarrow x \equiv 15 \pmod{17}$

Division by 2 is allowed since $\gcd(2, 17) = 1$.

Division:- The general rule is that you can divide by a (mod n) when $\gcd(a, n) = 1$.

Ex- Solve $5x + 6 \equiv 13 \pmod{11}$

$5x \equiv 7 \bmod 11$.

If we divide by 5 then what is $7/5$?

but we know $7 \bmod 11$ is same as $18 \equiv 29 \equiv 40 \equiv \dots \pmod{11}$

So we can write $7 \bmod 11$ as $40 \bmod 11$.

So $5x \equiv 40 \bmod 11$

Hence. $x \equiv 8 \bmod 11$

II method.

$\therefore 5 \cdot 9 \equiv 1 \pmod{11}$

9 is the multiplicative inverse of 5.

So Instead of dividing by 5 we can multiply by 9 Hence.

$5x \equiv 7 \pmod{11}$

$45x \equiv 63 \bmod 11$

$x \equiv 8 \bmod 11$

Ex. Solve. $11111x \equiv 4 \pmod{12345}$

Refer ~~Ex~~ $ax_1 + by_1 = \gcd(11111, 12345) = 1$

$11111x_1 + 12345y_1 = 1$

$11111 \cdot 2471 + 12345y_1 = 1$

So, ~~inverse~~ multiplicative inverse of 11111 is 2471. multiply both sides by 2471.

$11111 \cdot 2471x \equiv 9884 \pmod{12345}$

$x \equiv 9884 \pmod{12345}$

Finding $a^{-1} \pmod n$

(i) Use the extended Euclidean algorithm to find integers s and t such that

$as + nt = 1$, a & n are relatively prime.

(ii) $\bar{a} \equiv s \pmod n$. $as \equiv 1 \pmod n$

Ex. Solve $12x \equiv 21 \pmod{39}$, $\gcd(12, 39) = 3$

So $4x \equiv 7 \pmod{13}$, $\gcd(4, 13) = 1$

$4 \cdot x + 13 \cdot y_1 = 1$

So $x_1 =$ values 5, 18, 31 ...

$4 \cdot 5x \equiv 7 \cdot 5 \pmod{13}$

$4 \cdot 5 \cdot x \equiv 7 \cdot 5 \pmod{13}$

$7x \equiv 7 \cdot 5 \pmod{13}$

$x \equiv 5 \pmod{13}$

$12 \cdot 5x \equiv 21 \cdot 5 \pmod{39}$

$60x \equiv 21 \cdot 5 \pmod{39}$

$21x \equiv 21 \cdot 5 \pmod{39}$

$x \equiv 5 \pmod{39}$

$4x \equiv 7 \pmod{13}$

for $x = 5$

$4 \cdot 5 \equiv 7 \pmod{13}$

$20 \equiv 7 \pmod{13}$

$13 \equiv 0 \pmod{13}$

$4 \cdot 5x \equiv 7 \cdot 5 \pmod{13}$

$20x \equiv 35 \pmod{13}$

$7x \equiv 9 \cdot 5 \pmod{13}$

$35x \equiv 9$

The Chinese Remainder Theorem

In many situations, it is useful to break a congruence mod n into a system of congruences mod factors of n.

Ex. Let a number x satisfy $x \equiv 25 \pmod{42}$

So $x = 25 + 42k$ for some integer k. $42 = 7 \times 6$

$x = 25 + 7(6k) \Rightarrow x \equiv 25 \equiv 4 \pmod 7$

$x = 25 + 6(7k) \Rightarrow x \equiv 25 \equiv 1 \pmod 6$ Therefore

$$x \equiv 25 \pmod{42} \Rightarrow \begin{cases} x \equiv 4 \pmod 7 \\ x \equiv 1 \pmod 6 \end{cases}$$

This process can be reversed. A systems of congruences can be replaced by a single congruence under certain conditions.

Theorem - Suppose $\gcd(m, n) = 1$. Given integers a and b, there exists exactly one solution x (mod mn) to the simultaneous congruences

$$x \equiv a \pmod m, \quad x \equiv b \pmod n$$

Ex: Solve $x \equiv 3 \pmod 7$, $x \equiv 5 \pmod{15}$

Sol. → Since $80 \equiv 3 \pmod 7$ and $80 \equiv 5 \pmod{15}$ 80 is a solution.

$x \equiv 80 \pmod{105}$, $\begin{cases} 105 = 7 \times 15 \\ \gcd(7, 15) = 1 \end{cases}$

How do we get 80 ?

Method-I (for small Nos).

List the numbers congruent to b (mod n) until you find one that is congruent to a (mod m).

Ex. Numbers congruent to 5 (mod 15) are

5, 20, 35, 50, 65, 80, 95, ---

mod 7, these are 5, 6, 0, 1, 2, 3, 4, ---

$x \equiv 80 \equiv 3 \pmod 7$ also

$x \equiv 80 \equiv 5 \pmod{15}$

Hence $x \equiv 80 \pmod{105}$

Method-II - (for large Nos.)

For large Nos m, n

The numbers congruent to b (mod n) are of the form $b + nk$ with k an integer. So, we need to solve $b + nk \equiv a \pmod m$.

$$nk \equiv (a-b) \pmod m$$

Since $\gcd(m, n) = 1$. there is a multiplicative inverse i for n (mod m). Multiplication by i gives

$$k \equiv (a-b)\, i \pmod m$$

Substituting back into $x \equiv b + nk$, then reducing mod mn, gives the answer.

Ex. Solve $x \equiv 7 \pmod{12345}$, $x \equiv 3 \pmod{11111}$

Sol. From previous example 11111 $x \equiv 4 \pmod{12345}$ the inverse of $11111 \pmod{12345}$ is $i = 2471$.

Therefore, $k \equiv 2471(7-3) \equiv 9884 \pmod{12345}$.

This yields $x \equiv (3 + 11111 \cdot 9884) \equiv 109821137 \pmod{11111 \cdot 12345}$

Chinese Remainder Theorem (General)

2 → $m_1, m_2, \ldots m_k$ be integers with $\gcd(m_i, m_j) = 1$ whenever $i \neq j$. Given integers $a_1 \ldots a_k$, there exists exactly one solution $x \pmod{m_1 \ldots m_k}$ to the simultaneous congruences

$x \equiv a_1 \pmod{m_1}$, $x \equiv a_2 \pmod{m_2}$ ... $x \equiv a_k \equiv \pmod{m_k}$.

$$m = \prod_{j=1}^{k} m_j$$

Ex. $m = m_1 \cdot m_2 \cdot m_3 = 105$, with $m_1 = 3, m_2 = 5, m_3 = 7$.
$a_1 = 2, a_2 = 2, a_3 = 3$

Then least multiplicative inverse of $M_1 = m/m_1$
$= 35 \pmod{m_1 = 3}$ is $N_1 = 2$

The least multiplicative inverse of $M_2 = m/m_2$
$= 21 \bmod m_2 = 5$ is $N_2 = 1$ and

Least multiplicative inverse of $M_3 = m/m_3$
$= 15$ modulo $m_3 = 7$ is $N_3 = 1$ Then

$$x = \sum_{j=1}^{3} a_j N_j M_j = 2 \cdot 2 \cdot 35 + 2 \cdot 1 \cdot 21 + 3 \cdot 1 \cdot 15$$
$$= 227$$

By reducing $x = 227 \bmod N = 105$, we get $x = 17$.
Hence.

for $n = 17$.

$n \equiv a_1 \pmod{m_1} \equiv a_2 \pmod{m_2} \equiv a_3 \pmod{m_3}$.

$2 \pmod 3 \equiv 2 \pmod 5 \equiv 3 \pmod 7 \equiv 22 \pmod{105}$

or $2 \bmod 3 \equiv 2 \bmod 5 \equiv 3 \bmod 7 \equiv 17 \bmod 105$